

PROJECT INTERMIX

Sovereign Clarity + Continuity + Resonance

Release state

Resonance v1.3.1 completes the verified local voice path: Debian renders Kokoro audio, Termux validates it, and Android plays it through its native media route.

SHA-256 7fadfda85185643300edb723acad28317b498cd7d5aa47ab09261f8cba1056a1

PIXEL 10 PRO	8K LOCAL CONTEXT	PRIVATE SQLITE
Tensor G5 / Android 17	Adaptive virtual retention	Sessions, facts, FTS5

RELEASE LINE

VERSION	CODENAME	FOUNDATION
1.0.0	Memory	SQLite memory, migration, retrieval, prompt budgets
1.1.x	Horizon	Resident LiteRT engine and responsive cockpit
1.2.0	Continuity	Mission ledger, repair epochs, Workspace Lens
1.2.1	Clarity	Verified recall and final Markdown hierarchy
1.3.1	Resonance	Kokoro bridge, Android playback, Voice Archive

Purpose: preserve enough architecture, operating policy, installation detail, and verified state for a new development environment to continue safely without replaying the full build conversation.

Current system state

A stable developer build for daily local use; not yet an official distributable Android product.

What works now

- + Private, persistent SQLite conversation history and durable memory.
- + Cross-session recall backed by verified controller records.
- + Resident Gemma 4 E4B inference on LiteRT GPU with bounded 8K KV.
- + Adaptive response depth and prompt compaction instead of raw context growth.
- + Forced web grounding with fail-closed evidence guards.
- + Fixed-folder code generation, execution, repair, checkpoints, and reviewed deletion.
- + Kokoro voice generation in Debian with native Android playback from Termux.

Operating principles

- + Local-first: core chat, memory, code, and voice remain on device.
- + Truth before fluency: forced grounding never invents a result.
- + Controller authority: verified state outranks model narration.
- + Bounded autonomy: writes and tests are automatic only inside the fixed workspace.
- + Recoverability: snapshots, checkpoints, trash, and state ledgers precede mutation.
- + Visible work: long-running actions expose status and cancellation.

Proven device profile

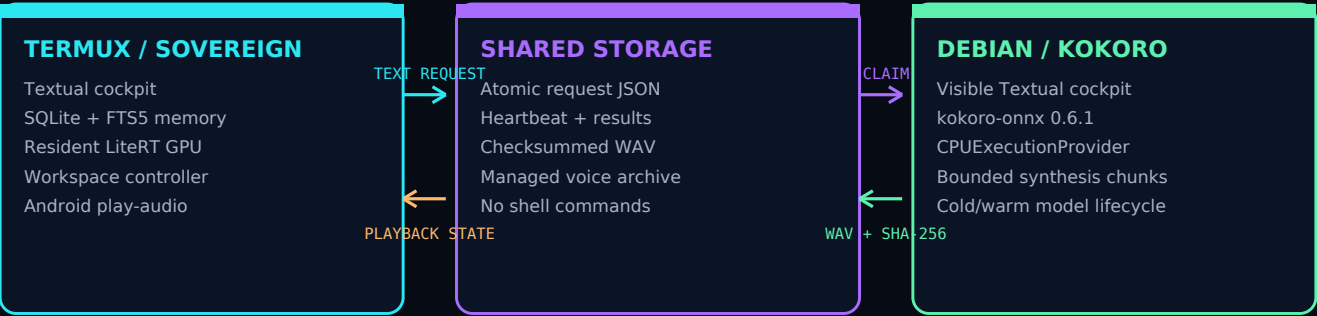
LAYER	VERIFIED CONFIGURATION
Device	Google Pixel 10 Pro / Tensor G5 / Android 17
Primary shell	Termux / Python 3.13 / Textual 8.2.8
Inference	LiteRT-LM 0.16.1 / Gemma 4 E4B / OpenCL GPU
Context	8,000-token KV cache; prompt ceiling varies by response mode
Voice VM	Debian / Python 3.13.5 / kokoro-onnx 0.6.1 / CPU provider
Audio	Termux play-audio 0.6-3; Debian Pulse auto_null deliberately bypassed
Thermal	External Peltier cooling; observed sustained operating target near 25 C
Storage	Project, models, SQLite, workspace, archives, and shared bridge state

FIELD NOTE

Cold-start latency and low-memory unloading are expected behaviors, not data loss. Sessions, facts, missions, and files are committed outside the model KV cache and rehydrate after restart.

Two runtimes, one local system

Termux owns intelligence and Android playback. Debian owns Kokoro synthesis. Shared storage is a narrow, validated data bridge.



Trust boundary: shared state is data only; each side validates path, state, and ownership.

Primary component map

MODULE	RESPONSIBILITY
tui_app.py	Textual cockpit, chat stream, Markdown finalization, telemetry, Workspace Lens, Voice Archive
llm_controller.py	Prompt assembly, memory routing, grounding policy, resident inference, hidden protocol filtering
memory_store.py + retriever	SQLite schema, FT55, facts, sessions, messages, provenance, expiration, retrieval
resident_engine.py	One hot LiteRT Engine, fresh stateless conversations, pressure and idle unload guards
agent.py + task_ledger.py	Ordered workspace tools, real execution results, repair epochs, verified completion
web_search.py	Multi-provider evidence retrieval, relevance and freshness guards, cache, cooldown
audio_bridge.py	Atomic request/result state, SHA-256 checks, playback ownership, archive rotation
kokoro_runtime.py	Lazy Kokoro load, bounded chunks, cancellation, atomic WAV output

OWNERSHIP RULE

Model output can propose actions, but only deterministic controller code reads the tags, enforces paths, performs mutations, records results, and decides whether a task is complete.

Finite context, durable continuity

The system gives an 8K model the experience of a longer working memory by selecting state, not by expanding the physical cache.

IDENTITY + POLICY	Always retained
CURRENT REQUEST	Never compacted
DURABLE FACTS	Ranked by relevance and priority
SESSION CHECKPOINT	Rolling compressed continuity
RECENT TURNS	Recency slice
MISSION LEDGER	Workspace mode only

Raw transcripts remain in SQLite even when they are not selected into the next prompt.

Storage classes

- + Raw messages: complete transcript in SQLite.
- + Sessions: restartable chat timelines and titles.
- + Durable facts: explicit user facts, preferences, and project knowledge.
- + Pinned memory: high-priority facts retained across prompts.
- + Provenance: source-backed hard facts carry retrieval time and expiry.
- + Mission records: exact files, hashes, tests, outputs, failures, and completion.
- + Rolling checkpoint: compact session summary for older turns.

Adaptive context allocation

MODE	INPUT CEILING	OUTPUT RESERVE	VISIBLE TARGET
Brief	7,200	800	Up to 5 sentences
Normal	6,800	1,200	Up to 10 sentences
Deep	5,800	2,200	Request-sized explanation
Workspace	5,200	2,800	Tools, feedback, repair, verification

Privacy and revision policy

INPUT	DURABLE ACTION	GUARD
Explicit preference	Eligible for durable memory	Inspectable, pinnable, forgettable
Sensitive inference	Rejected	No inferred diagnosis or mental-health label
Source-backed hard fact	Requires matching provenance	Expires or revises when evidence changes
Conflicting memory	Prior revision preserved	Newest verified state becomes active
Old conversation	Compacted for prompts	Full transcript remains queryable

Resident performance with pressure guards

A long-lived GPU Engine removes repeated model initialization, while fresh conversations prevent KV state from accumulating indefinitely.

STATE	MEANING	ACTION
COLD	Engine absent	Initialize on the next request
WARMING	LiteRT loads model and OpenCL delegates	Show progress; do not freeze input handling
HOT	Engine resident	Create a fresh stateless conversation per assembled prompt
UNLOADING	Idle or RAM pressure threshold reached	Close Engine and recover memory
RECOVERY	Process restarted or Android killed it	Rehydrate from SQLite and mission state

Measured smoke test

On the Pixel 10 Pro, the persistent GPU Engine initialized in approximately 6.55 seconds. The first text arrived about 1.21 seconds after initialization, and the complete minimal test ended in 8.99 seconds.

Longer real prompts can still spend noticeably more time in prompt assembly and prefill. Field observations ranged from roughly 15 to 50 seconds before first visible text on cold or larger contexts.

Memory policy

- + Keep the engine hot while the cockpit is active and headroom is healthy.
- + Unload after sustained inactivity or low available memory.
- + Treat Android SIGKILL as a process failure, not a session reset.
- + Never depend on KV state as the only copy of project progress.

Latency interpretation

PHASE	COST	OPTIMIZATION
Context assembly	SQLite retrieval, mission record, policy, optional web evidence	Reduce normal-chat selection before shrinking the 8K ceiling
Engine initialization	Cold model and GPU delegate setup	Resident reuse or optional warm-up after typing begins
Prefill	Processing selected prompt tokens	Normal mode should avoid dragging in irrelevant history
Generation	Token decode and UI repaint	Throttle repaint; render Markdown once at completion
Memory tail	Hidden extraction and persistence	Emit only changed memory fields and keep protocol hidden

THERMAL NOTE

External cooling helps sustain clocks, but it does not prevent Android low-memory management. Available RAM remains the decisive safety signal.

Agentic work with a verified ledger

Intermix can read, write, and test automatically inside one fixed folder. Deletion remains reviewable and recoverable.

Fixed scope

~/storage/downloads/sovereign_workspace/

CAPABILITY	DEFAULT	ENFORCEMENT
Read	Automatic	Normalized paths must remain inside workspace
Write	Automatic	Atomic replacement plus content-addressed checkpoint
Run Python	Automatic	Native Termux process, fixed cwd, bounded environment and timeout
Cancel	Immediate request	Stops current process and discards remaining generated actions
Delete	Explicit approval	Hash revalidated, then moved to recoverable trash
Complete	Controller-owned	Requires a successful test after the latest write

Repair epoch

- + Model proposes ordered tool actions.
- + Controller executes one action at a time.
- + A failure stops later speculative actions.
- + Real stdout, stderr, exit code, and file hash enter the mission ledger.
- + A fresh inference sees the verified failure and proposes the next repair.
- + Completion is accepted only after post-write verification.

Continuity surfaces

- + Task state survives fresh model conversations and application restart.
- + Ctrl+E opens Workspace Lens; Ctrl+S saves atomically.
- + Changed files receive checkpoints beneath .intermix/checkpoints.
- + Approved deletions move to .intermix/trash.
- + PROJECT_STATE.md updates only when structural hashes change.
- + Raw XML and generated file bodies stay out of normal chat.

BOUNDARY

This is a controlled native-code workspace boundary, not an Android OS sandbox. The path fence, timeout, approval model, checkpoints, and visible cancellation are the real safeguards.

Fresh information without confident invention

Search is lightweight for the phone. Relevance, provenance, and failure behavior matter more than local compute.

Two different web paths

PATH	TRIGGER	FAILURE CONTRACT
/web	Always attempts retrieval	Visible failure; inference stops when no verifiable evidence exists
Automatic grounding	Only true external and time-sensitive lookup intent	Quiet miss; model is constrained from making current claims
Idle research scout	Only while TUI is open and idle	Read-only findings from configured trusted source classes

Evidence pipeline

- + Classify volatility and lookup intent.
- + Route to query-specific providers.
- + Retrieve bounded candidates and page evidence.
- + Reject related but nonresponsive results.
- + Apply freshness rules for current-update claims.
- + Cache accepted evidence with retrieval time and expiry.
- + Require citations before showing a forced-web answer.

Trusted idle sources

- + Official repositories and documentation.
- + Verified registries and release metadata.
- + Community issues and discussions as unverified leads.
- + General web results disabled for autonomous research.
- + Findings can inform proposals but never execute instructions.

The scout waits for 60 idle seconds, spaces maintenance passes, limits requests, and skips research when free storage is below the configured safety floor.

Known limitation

Provider availability and search relevance remain inconsistent. DuckDuckGo can rate-limit quickly, while generic Wikipedia matches may be valid background but insufficient evidence for a current device update. The guard refusing a weak result is correct behavior, even when it feels less seamless than a cloud assistant.

Recommended next search increment

- + Add official-source resolvers for Android, Pixel, GitHub releases, PyPI, and other query classes.
- + Fetch a bounded authoritative page after search rather than trusting snippets alone.
- + Keep DuckDuckGo as a fallback, with shared cooldown across its routes.
- + Measure provider coverage and answer quality separately from inference latency.

Sovereign Horizon cockpit

The visual system favors hierarchy, concentration, and Android desktop-mode readability over terminal spectacle.

SURFACE	DESIGN CONTRACT
Palette	Near-black navy, cyan structure, purple interaction, white/grey generated text
Streaming	Plain text repainted at a bounded interval instead of every token
Completion	One final native Markdown render for headings, lists, emphasis, code, and quotes
Diagnostics	Long local-system and sandbox output collapses; warnings remain visible
Sidebar	Model, release, context, memory, mission, engine, storage, and voice telemetry
Responsive layout	Desktop-first two-column layout with compact behavior for narrow terminals
Typography	Terminal cells cannot vary real font size; hierarchy uses weight, color, spacing, and panels

Core and memory

```
/status
/engine status
/inspect context
/memory facts
/memory search <topic>
/sessions
/new [title]
/export
```

Workspace

```
/workspace <goal>
/files
/task status
/tasks
/approvals
/approve <id>
/deny <id>
/cancel
```

Depth and research

```
/brief <message>
/normal <message>
/deep <message>
/response status
/web <query>
/research status
/research findings
/docs refresh
```

Keyboard controls

KEY	ACTION	KEY	ACTION
Ctrl+E	Workspace Lens	Ctrl+S	Save selected workspace file
Ctrl+V	Voice Archive	Ctrl+I	Pin or unpin selected WAV
Ctrl+X	Cancel active tool or voice playback	Ctrl+Q	Exit cockpit safely

ANDROID UI NOTE

The Termux extra-keys bar is owned by Termux, not Textual. A future convenience patch can switch profiles or document the Termux property change, but the cockpit should not silently rewrite global keyboard settings.

Verified local speech bridge

Speech is generated only from a completed, protocol-filtered response and begins only after an explicit VOICE action.

STATE	MEANING	AUTHORITY
READY	A completed visible response is available	VOICE may be pressed once
TRANSMITTING	Termux atomically publishes checksummed text	Debian may claim it
RENDERING	Kokoro is producing bounded chunks	Cancellation remains available
AUDIO_READY	WAV and SHA-256 are committed	Termux validates path and hash
STARTING	Termux owns the exclusive playback claim	No other request may start
PLAYING	Android play-audio process is active	Ctrl+X requests cancellation
COMPLETED	Player exited successfully	Recording enters archive rotation
FAILED / CANCELLED	Truthful terminal state	WAV remains available for inspection

Why Termux plays

The Debian VM exposed only PulseAudio's silent `auto_null` sink. The WAV itself was valid: 24 kHz audio with measurable peak and RMS levels. Termux `play-audio` produced audible Android media output and returned `AUDIO_PROBE_OK`. Therefore Debian owns synthesis; Termux owns final playback and truthfully reports the player lifecycle.

Archive policy

- + Keep the newest 25 unpinned managed recordings.
- + Never rotate queued, starting, playing, or pinned files.
- + Ignore unrelated WAVs and files outside the managed archive.
- + Clean abandoned `.part` files only after one hour.
- + Inspect duration, sample rate, channels, size, and pin state read-only.

Pinned or active recordings may place the physical total above 25.

Kokoro defaults

SETTING	VALUE	SETTING	VALUE
Voice	af_heart	Speed	1.0
Autoplay in Kokoro cockpit	true	Kokoro idle unload	300 seconds
Text ceiling	20,000 characters	Bridge heartbeat	5-10 second UI polling

09 / RECOVERY

Install and rollback reference

Install Debian first, reopen its cockpit, then patch Termux. Both installers test before copying and snapshot before mutation.

Release artifact

```
Project_Intermix_Resonance_v1.3.1.zip
SHA-256 7fadfda85185643300edb723acad28317b498cd7d5aa47ab09261f8cba1056a1
```

1. Debian Terminal

```
resonance_zip="/mnt/shared/Project_Intermix_Resonance_v1.3.1.zip"
test -f "$resonance_zip"
test "$(sha256sum "$resonance_zip" | cut -d' ' -f1)" =
    "7fadfda85185643300edb723acad28317b498cd7d5aa47ab09261f8cba1056a1"
installer="$HOME/intermix-resonance-131-debian"
mkdir -p "$installer"
python -m zipfile -e "$resonance_zip" "$installer"
cd "$installer/Project_Intermix_Resonance_v1.3.1"
bash ./install_debian.sh "$HOME/kokoro_tts" "$HOME/kokoro-env"
```

2. Termux

```
resonance_zip="$HOME/storage/shared/Project_Intermix_Resonance_v1.3.1.zip"
test -f "$resonance_zip"
test "$(sha256sum "$resonance_zip" | cut -d' ' -f1)" =
    "7fadfda85185643300edb723acad28317b498cd7d5aa47ab09261f8cba1056a1"
installer="$HOME/intermix-resonance-131-termux"
mkdir -p "$installer"
python -m zipfile -e "$resonance_zip" "$installer"
cd "$installer/Project_Intermix_Resonance_v1.3.1"
bash ./install_termux.sh "$HOME/project-intermix"
sovereign
```

Rollback locations

```
~/kokoro_tts/archive/pre_resonance_131_debian_*
~/project-intermix/archive/pre_resonance_131_termux_*
```

Restore only while the affected cockpit is closed. The Debian uninstaller removes managed cockpit code and its Bash block while preserving models, voices, configuration, generated audio, speak.py, and bridge state.

Acceptance and daily testing

The packaged release passed isolated Resonance tests, a fresh extracted-package run, and the inherited Clarity regression suite.

SCOPE	RESULT	EVIDENCE
Resonance package	20 tests	Playback success/failure, SHA-256, path guards, stale recovery, archive rotation, pinning, lazy Kokoro, install patches
Clarity regression	34 tests	Memory, FTS5, prompt budgets, recall, grounding, workspace repair, UI rendering, research policy
Python compile	Pass	All shipped Python modules compiled
Shell syntax	Pass	Debian and Termux installers validated
Extracted ZIP	Pass	Version, tree, checksum, and tests reverified after packaging
Device audio probe	AUDIO_PROBE_OK	Termux Android playback audible from a Kokoro-generated WAV

Post-install acceptance

- + Debian Kokoro Cockpit autostarts with MODEL COLD.
 - + Sovereign reports VOICE: ONLINE within one heartbeat interval.
 - + VOICE stays disabled until a response is final and protocol-filtered.
 - + Activity advances through transmitting, rendering, starting, playing, and completed.
 - + Audio is audible through Android and completion appears only after playback ends.
- + Ctrl+V shows the new WAV; Ctrl+I toggles pin state.
 - + Ctrl+X ends a longer narration as cancelled, not completed.
 - + Closing Debian changes voice to OFFLINE while local chat remains usable.
 - + Reopening Debian restores ONLINE without restarting Sovereign.
 - + Restarting both cockpits rehydrates sessions, memory, missions, workspace, and draft independently.

Daily field log - minimum useful data

AREA	RECORD
Chat	Cold/hot first-token delay; selected prompt tokens; output length
Memory	Correct recall, incorrect recall, unwanted durable fact, conflict handling
Web	Query, provider, accepted/refused, source relevance, elapsed time
Workspace	Goal, actions, failure, repair count, test result, cancellation behavior
Voice	Cold/warm render time, playback state accuracy, cancellation, archive count
Resources	Available RAM before/after engine, thermal state, storage remaining

Recommended next increments

Sustain daily use before expanding the autonomy surface. Measured friction should choose the next release.

ORDER	INCREMENT	BOUNDARY
1	Stability telemetry	Separate context assembly, engine init, prefill, first token, decode, memory tail, and TTS timings
2	Official-source grounding	Query adapters and bounded authoritative page fetches; DuckDuckGo only as fallback
3	Proposal-only diagnostics	Core may present evidence, hypothesis, runtime, files affected, and requested approval
4	Resource recommendations	Turn RAM, storage, thermal, and latency measurements into visible non-destructive advice
5	Preference weighting	Priority, confidence, revision history, and task-specific tone calibration
6	Tool orchestration	Per-tool permissions, runtime budgets, acceptance criteria, and auditable chains
7	Multimodal pipeline	Only after text, memory, grounding, workspace, and voice remain stable in daily use
8	Official deployment	Signed device profiles, resumable checkpoints, atomic versions, cache warming, responsive presets

Deliberately deferred

- + Automatic Android Terminal launch.
- + Narration without a user click.
- + Streaming partial-sentence speech.
- + Unsolicited diagnostics or fixes.
- + General-web autonomous research.
- + Application-level multimodal input.
- + Official consumer deployment claims.

Resume checklist

- + Read this handoff and the release README.
- + Confirm VERSION and ZIP SHA-256.
- + Inspect the newest rollback snapshots.
- + Run package tests before installing a patch.
- + Preserve SQLite, workspace .intermix state, and Kokoro assets.
- + Use controller records as truth for completed work.
- + Treat model-generated roadmaps as proposals, not authority.

Canonical paths

ASSET	PATH	ASSET	PATH
Termux	~/project-intermix/	Workspace	~/storage/downloads/sovereign_workspace/
SQLite	~/project-intermix/memory/sovereign.db	Kokoro	/home/droid/kokoro_tts/
Model	~/project-intermix/models/gemma-4-E4B-it.litertlm	Voice config	~/config/intermix-kokoro/config.json

HANDOFF STATE

Project Intermix Resonance v1.3.1 is installed, audible, daily-usable, memory-safe by design, and ready for sustained field testing. The correct next action is observation, not feature inflation.